Static Bulk Dielectric Response from the Modern Theory of Polarization

A self-consistent finite-electric-field workflow in plane-wave DFT

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Objective. For an insulating bulk crystal, obtain (i) the *ion-clamped electronic* dielectric tensor $\boldsymbol{\epsilon}_\infty$ and (ii) the *relaxed-ion static* dielectric tensor $\boldsymbol{\epsilon}_0$ using Berry-phase polarization of self-consistent field-polarized occupied Bloch states. The lattice contribution is $\boldsymbol{\epsilon}_{\text{latt}} = \boldsymbol{\epsilon}_0 - \boldsymbol{\epsilon}_\infty$. The same polarization can be interpreted through occupied Wannier centers. All dielectric tensors below are relative tensors, with Gaussian/atomic-unit conventions unless noted.

1 Physical setting and definitions

Consider an insulating periodic crystal with primitive-cell volume Ω, ionic coordinates $\{\mathbf{R}_I\}$, and orthonormal occupied Bloch functions

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \qquad u_{n\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r}). \tag{1}$$

The imposed field $\boldsymbol{\mathcal{E}}$ denotes the *total macroscopic electric field inside the crystal*, held fixed by the electrical boundary conditions; it should not be confused with the unscreened vacuum field of fixed external free charges. The electronic structure responds self-consistently at fixed $\boldsymbol{\mathcal{E}}$. The displacement field is $\mathbf{D} = \boldsymbol{\mathcal{E}} + 4\pi\mathbf{P}$ (Gaussian units).

The polarization susceptibility and dielectric tensor are

$$\chi_{\alpha\beta} = \left.\frac{\partial P_\alpha}{\partial \mathcal{E}_\beta}\right|_{\boldsymbol{\mathcal{E}}=0}, \qquad \epsilon_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi\chi_{\alpha\beta}. \tag{2}$$

Keeping the lattice and ions fixed defines $\boldsymbol{\epsilon}_\infty$ (the static electronic or "high-frequency" tensor); relaxing the internal coordinates at fixed strain defines $\boldsymbol{\epsilon}_0$. This is a static, long-wavelength *macroscopic* response, not the full microscopic $\epsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q},\omega)$.

2 Why electric enthalpy is the variational functional

For a finite neutral system, the electric dipole moment is $\boldsymbol{\mu} = e\sum_I Z_I\mathbf{R}_I - e\sum_i \mathbf{r}_i$, with Z_I the ionic-core charges appropriate to the electron–ion treatment. Its coupling to a prescribed uniform field is $-\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mu}$. Identifying $\boldsymbol{\mu} = \Omega\mathbf{P}$ per bulk cell motivates the *electric enthalpy* [2]:

$$\boxed{\mathcal{F}[\{u_{n\mathbf{k}}\},\{\mathbf{R}_I\};\boldsymbol{\mathcal{E}}] = E_{\text{DFT}}[\{u_{n\mathbf{k}}\},\{\mathbf{R}_I\}] - \Omega\boldsymbol{\mathcal{E}}\cdot\mathbf{P}[\{u_{n\mathbf{k}}\},\{\mathbf{R}_I\}].} \tag{3}$$

Here E_{DFT} is the *total zero-field DFT energy functional evaluated on the field-polarized state*, not the sum of occupied Kohn–Sham eigenvalues. It contains the kinetic, electron–ion, Hartree, exchange–correlation, ion–ion, and double-counting terms appropriate to the chosen DFT method.

Equivalently, if $U(\mathbf{P})$ is the constrained internal energy at fixed polarization on a specified insulating branch, switching to prescribed $\boldsymbol{\mathcal{E}}$ gives the Legendre transform $\mathcal{F}(\boldsymbol{\mathcal{E}}) = \text{stat}_{\mathbf{P}}[U(\mathbf{P}) - \Omega\boldsymbol{\mathcal{E}}\cdot\mathbf{P}]$. A homogeneous scalar potential $\phi = -\boldsymbol{\mathcal{E}}\cdot\mathbf{r}$ is not lattice periodic; the modern theory replaces the ill-defined bulk dipole expectation value with a Berry phase of occupied periodic Bloch functions.

At nonzero field the infinite crystal has no globally stable insulating ground state: intercell charge transfer can lower its energy indefinitely. On a finite $\mathbf{k}$ mesh, however, sufficiently weak fields support insulating *local* minima continuously connected to the zero-field state; the computational task is to follow that metastable branch, not to seek an unrestricted global minimum [2].

3 Berry-phase polarization and Wannier centers

3.1 Continuum expression

The total polarization is the sum of electronic and ionic terms,

$$\mathbf{P} = \mathbf{P}_{\mathrm{el}} + \mathbf{P}_{\mathrm{ion}}, \tag{4}$$

$$\mathbf{P}_{\mathrm{el}} = -\frac{e}{(2\pi)^3} \sum_{n\in\mathrm{occ}} \int_{\mathrm{BZ}} d^3k \; \mathbf{A}_n(\mathbf{k}), \qquad \mathbf{A}_n(\mathbf{k}) = i\langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} u_{n\mathbf{k}} \rangle, \tag{5}$$

$$\mathbf{P}_{\mathrm{ion}} = \frac{e}{\Omega} \sum_I Z_I \mathbf{R}_I. \tag{6}$$

The sum in Eq. (5) counts individual occupied spin states; in a spin-degenerate calculation one can instead sum spatial orbitals with occupation factor two. For an entangled occupied manifold, the Berry connection is understood as the trace over the full occupied subspace. Polarization is multivalued:

$$\mathbf{P} \;\equiv\; \mathbf{P} + \frac{e\mathbf{R}}{\Omega}, \qquad \mathbf{R} \text{ any Bravais lattice vector.} \tag{7}$$

Therefore, all field-induced polarization differences must be compared on a *continuously unwrapped branch* [1].

3.2 Practical discrete formula for plane-wave states

On a uniform mesh, construct occupied-band overlap matrices along a reciprocal-space string in lattice direction i:

$$M^{(j)}_{mn} = \langle u_{m,\mathbf{k}_j} \,|\, u_{n,\mathbf{k}_{j+1}} \rangle, \qquad \varphi_i = -\,\mathrm{Im}\ln \prod_{j\in\mathrm{string}} \det M^{(j)}. \tag{8}$$

Adjacent-string end points are identified using the appropriate reciprocal-lattice boundary phase. Average φ_i over the transverse strings, unwrapping their phases consistently. For primitive vectors $\mathbf{a}_i$, the electronic polarization can be assembled from the resulting $\bar{\varphi}_i$ as

$$\mathbf{P}_{\mathrm{el}} = -\frac{e}{2\pi\Omega} \sum_{i=1}^{3} \mathbf{a}_i \, \bar{\varphi}_i \qquad (\text{mod polarization quanta}). \tag{9}$$

This is a compact notation for a regular-mesh Berry-phase computation; spin occupation factors, $\mathbf{k}$-point weights, and overlap augmentation for ultrasoft/PAW methods must be treated consistently in an implementation.

3.3 Equivalent occupied-Wannier-center description

When the occupied manifold admits a smooth localized Wannier gauge, with home-cell centers $\bar{\mathbf{r}}_n = \langle w_{n\mathbf{0}} | \mathbf{r} | w_{n\mathbf{0}} \rangle$, the same result is

$$\boxed{\mathbf{P}_{\mathrm{el}} = -\frac{e}{\Omega} \sum_{n\in\mathrm{occ}} \bar{\mathbf{r}}_n \quad \left(= -\frac{2e}{\Omega} \sum_{n\in\mathrm{occ,spatial}} \bar{\mathbf{r}}_n \text{ for spin degeneracy} \right).} \tag{10}$$

Thus the field-induced electronic polarization can be understood as the net shift of occupied Wannier centers:

$$\Delta\mathbf{P}_{\mathrm{el}} = -\frac{e}{\Omega} \sum_n \left[\bar{\mathbf{r}}_n(\boldsymbol{\mathcal{E}}) - \bar{\mathbf{r}}_n(\mathbf{0}) \right] \quad \text{at fixed cell volume and a continuous branch.} \tag{11}$$

Individual centers and their orbital-resolved contributions depend on the Wannier gauge; the *total* is gauge invariant modulo a polarization quantum. Maximally localized Wannier functions are useful for interpretation but are *not required* to compute Eq. (9). Wannierizing a disentangled set containing empty bands is not a replacement for evaluating the full occupied-manifold polarization.

4 Self-consistent finite-field electronic calculation

For every small field $\boldsymbol{\mathcal{E}}$, use the converged zero-field orbitals as the initial state and locally minimize Eq. (3) under the occupied-orbital orthonormality constraints. The important self-consistent feedback is

$$\{u_{n\mathbf{k}}\} \longrightarrow n(\mathbf{r}) \longrightarrow V_H[n] + V_{\mathrm{xc}}[n] \longrightarrow \{u_{n\mathbf{k}}\}, \tag{12}$$

while the Berry-phase field term couples neighboring $\mathbf{k}$ points in the discretized functional. Thus the update is *not* a conventional independent-$\mathbf{k}$ diagonalization with a periodic sawtooth potential. The minimized result includes Hartree and exchange–correlation screening within the selected approximate DFT functional, including microscopic local-field feedback on the macroscopic response.

Algorithm A: clamped-ion tensor $\boldsymbol{\epsilon}_\infty$.

1. Relax the zero-field structure if needed. Fix the lattice vectors and ionic coordinates. Converge the insulating ground state $\{u^{(0)}_{n\mathbf{k}}\}$, density, total energy, and Berry-phase polarization $\mathbf{P}^{(0)}$.
2. For each Cartesian direction $\beta \in \{x, y, z\}$, choose a small field amplitude h_β and run separate self-consistent calculations for $\boldsymbol{\mathcal{E}} = \pm h_\beta \hat{\mathbf{e}}_\beta$, initialized from $\{u^{(0)}_{n\mathbf{k}}\}$.
3. At each converged field, evaluate the total Berry-phase polarization with fixed ions: $P_\alpha^{\pm\beta} = P_\alpha(\pm h_\beta \hat{\mathbf{e}}_\beta; \{\mathbf{R}_I^{(0)}\})$. Unwrap to the zero-field polarization branch.
4. Form the central finite-difference susceptibility and dielectric tensor:

$$\boxed{\chi^{\mathrm{el}}_{\alpha\beta} \simeq \frac{P_\alpha^{+\beta} - P_\alpha^{-\beta}}{2h_\beta}, \qquad \epsilon^{\infty}_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi\chi^{\mathrm{el}}_{\alpha\beta}.} \tag{13}$$

5. Repeat for smaller h_β and denser $\mathbf{k}$ meshes until the tensor is stable.

A central difference cancels the leading even-in-field errors in the polarization and works in polar as well as centrosymmetric crystals, provided the same branch is followed. The tensor may be computed in Cartesian coordinates without assuming cubic symmetry.

5 Ionic relaxation and the full static dielectric tensor

To include lattice screening at *fixed strain*, repeat the finite-field calculation allowing internal ionic coordinates to relax while retaining a self-consistent electronic minimization at each geometry. Denote the relaxed positions by $\mathbf{R}_I(\boldsymbol{\mathcal{E}})$, and calculate

$$\mathbf{P}^{\mathrm{rel}}(\boldsymbol{\mathcal{E}}) = \mathbf{P}_{\mathrm{el}}[\{u^{(\boldsymbol{\mathcal{E}})}_{n\mathbf{k}}\}, \{\mathbf{R}_I(\boldsymbol{\mathcal{E}})\}] + \frac{e}{\Omega}\sum_I Z_I \mathbf{R}_I(\boldsymbol{\mathcal{E}}). \tag{14}$$

Here the *electronic term also changes because the ions move*. The lattice contribution is not just the bare-ion displacement term in Eq. (6).

Algorithm B: relaxed-ion tensor $\boldsymbol{\epsilon}_0$.

1. Begin with the same zero-field equilibrium geometry and electronic state used for Algorithm A.
2. For every $\boldsymbol{\mathcal{E}} = \pm h_\beta \hat{\mathbf{e}}_\beta$, relax the internal ionic coordinates by minimizing the *same electric enthalpy* with respect to both ionic positions and occupied Bloch states; keep lattice vectors fixed.
3. Compute $\mathbf{P}^{\mathrm{rel}}(\pm h_\beta \hat{\mathbf{e}}_\beta)$ from the converged electronic Berry phase *plus* the ionic-core term. Unwrap polarization branches.
4. Extract
$$\boxed{\epsilon^0_{\alpha\beta} \simeq \delta_{\alpha\beta} + 4\pi \frac{P^{\mathrm{rel}}_\alpha(+h_\beta \hat{\mathbf{e}}_\beta) - P^{\mathrm{rel}}_\alpha(-h_\beta \hat{\mathbf{e}}_\beta)}{2h_\beta}.} \tag{15}$$
5. Subtract the tensors at matching convergence settings: $\boldsymbol{\epsilon}_{\mathrm{latt}} = \boldsymbol{\epsilon}_0 - \boldsymbol{\epsilon}_\infty$.

At fixed $\boldsymbol{\mathcal{E}}$ the electric-enthalpy force is

$$F_{I\beta} = -\frac{\partial \mathcal{F}}{\partial R_{I\beta}} = F^0_{I\beta} + \Omega \sum_\alpha \mathcal{E}_\alpha \frac{\partial P_\alpha}{\partial R_{I\beta}} = F^0_{I\beta} + e \sum_\alpha Z^*_{I,\alpha\beta} \mathcal{E}_\alpha, \tag{16}$$

where the Born effective charge Z^* in the last identity is evaluated self-consistently at the given structure and electrical conditions; the linear-response form uses the zero-field Z^*. Fully variational finite-field forces or an equivalent compatible force implementation are required for the ionic relaxation. If the cell strain is also relaxed, piezoelectric/elastic degrees of freedom contribute and the result is a *constant-stress*, rather than clamped-strain, dielectric response.

6 Equivalent small-displacement route: Born charges and force constants

The direct ionic relaxation can be cross-checked without nonzero-field ionic optimization. At zero field define the Born effective charges and force constants,

$$Z^*_{I,\alpha\beta} = \frac{\Omega}{e} \left. \frac{\partial P_\alpha}{\partial R_{I\beta}} \right|_{\boldsymbol{\mathcal{E}}=0} = Z_I \delta_{\alpha\beta} - \sum_{n \in \mathrm{occ}} \frac{\partial \bar{r}_{n\alpha}}{\partial R_{I\beta}}, \tag{17}$$

$$\Phi_{I\beta,J\gamma} = \left. \frac{\partial^2 E_{\mathrm{DFT}}}{\partial R_{I\beta} \partial R_{J\gamma}} \right|_{\boldsymbol{\mathcal{E}}=0}. \tag{18}$$

The second equality of Eq. (17) uses the individual-spin-state Wannier-center convention; for doubly occupied spatial Wannier orbitals the Wannier-center sum has a factor of two. The ionic contribution to the static response follows from the harmonic expansion

$$\mathcal{F} \simeq E_0 + \frac{1}{2} \sum_{I\beta,J\gamma} u_{I\beta} \Phi_{I\beta,J\gamma} u_{J\gamma} - e \sum_{I\alpha\beta} \mathcal{E}_\alpha Z^*_{I,\alpha\beta} u_{I\beta} - \Omega \boldsymbol{\mathcal{E}} \cdot \mathbf{P}_0 - \frac{\Omega}{2} \boldsymbol{\mathcal{E}}^{\mathsf{T}} \boldsymbol{\chi}^{\mathrm{el}} \boldsymbol{\mathcal{E}}. \tag{19}$$

Minimizing with respect to ionic displacements $u_{I\beta}$ gives $\mathbf{u} = e\Phi^+ Z^{*\mathsf{T}} \boldsymbol{\mathcal{E}}$. Consequently,

$$\boxed{\epsilon^0_{\alpha\gamma} = \epsilon^\infty_{\alpha\gamma} + \frac{4\pi e^2}{\Omega} \sum_{I\beta,J\delta} Z^*_{I,\alpha\beta} (\Phi^+)_{I\beta,J\delta} Z^*_{J,\gamma\delta}.} \tag{20}$$

The pseudoinverse Φ^+ removes the three acoustic translational zero modes. In a neutral insulating unit cell the Born-charge acoustic sum rule is $\sum_I Z^*_{I,\alpha\beta} = 0$. Soft infrared-active optical modes can give large positive lattice contributions. Equations (15) and (20) agree in the harmonic, weak-field limit when both calculations use matching DFT approximations, strain boundary conditions, and converged numerical settings.

7 Independent energy-curvature check

Let $\mathcal{F}_{\min}(\boldsymbol{\mathcal{E}})$ be the electric enthalpy evaluated along the locally stationary insulating branch. The envelope theorem implies

$$P_\alpha(\boldsymbol{\mathcal{E}}) = -\frac{1}{\Omega}\frac{\partial \mathcal{F}_{\min}}{\partial \mathcal{E}_\alpha}, \qquad \chi_{\alpha\beta} = -\frac{1}{\Omega}\left.\frac{\partial^2 \mathcal{F}_{\min}}{\partial \mathcal{E}_\alpha \partial \mathcal{E}_\beta}\right|_{\mathbf{0}}. \tag{21}$$

Near zero field, including any spontaneous polarization,

$$\mathcal{F}_{\min}(\boldsymbol{\mathcal{E}}) = \mathcal{F}_0 - \Omega \mathbf{P}_0 \cdot \boldsymbol{\mathcal{E}} - \frac{\Omega}{2}\boldsymbol{\mathcal{E}}^{\mathsf{T}}\boldsymbol{\chi}\boldsymbol{\mathcal{E}} + O(\mathcal{E}^3). \tag{22}$$

Thus the curvature of field-dependent electric enthalpy provides a secondary consistency check on the polarization finite differences. Use the clamped-ion branch for $\boldsymbol{\epsilon}_\infty$ and the relaxed-ion branch for $\boldsymbol{\epsilon}_0$. A polarization slope is often simpler to converge than a small difference of total energies.

8 Numerical precautions and validation

1. **Insulating branch.** Verify a nonzero gap and follow the field-polarized state continuously from zero field. Do not apply this ordinary static-insulator protocol to a metal or an unstable conducting branch.

2. **Field size and k mesh.** Test several symmetric field amplitudes; use the smallest fields that remain well above numerical polarization noise. The metastable field threshold decreases as the **k**-mesh density along the field direction increases [2]. Converge mesh and field *together*.

3. **Berry-phase branches.** Unwrap 2π phases between nearby fields/ionic geometries. A polarization quantum $e\mathbf{R}/\Omega$ must not be mistaken for an induced response.

4. **Occupied space and basis.** Include every occupied band and spin state in the Berry-phase overlap determinant. Ensure consistency of pseudopotential/PAW overlap operators, **k**-point weights, and convergence tolerances.

5. **SCF and forces.** Tighten electronic convergence sufficiently to resolve the small polarization differences. For relaxed-ion calculations also converge finite-field forces and ionic relaxation thresholds.

6. **Tensor checks.** For a nonmagnetic equilibrium material without nonreciprocal effects, the static dielectric tensor should be symmetric to numerical accuracy; also verify crystal-symmetry constraints. For a stable dielectric branch, its physical eigenvalues should be positive.

7. **Boundary conditions.** Record whether ions, cell shape, and strain were constrained. Fixed-$\boldsymbol{\mathcal{E}}$ and fixed-**D** are distinct electrical boundary conditions [3].

In SI units replace Eq. (2) with $\epsilon_{r,\alpha\beta} = \delta_{\alpha\beta} + \epsilon_{\text{vac}}^{-1}\,\partial P_\alpha/\partial \mathcal{E}_\beta$, using P in $\mathrm{C\,m^{-2}}$ and $\mathcal{E}$ in $\mathrm{V\,m^{-1}}$. The symbol ϵ_{vac} here is the vacuum permittivity, not the material static tensor $\boldsymbol{\epsilon}_0$.

9 Compact end-to-end implementation summary

Stage	Operation and output
0	Converge zero-field insulating bulk DFT, structure, Bloch states, and Berry-phase $\mathbf{P}(0)$.
1	For each $\beta = x, y, z$ and sign $\pm$, initialize the occupied states from the zero-field solution and locally minimize $E_{\mathrm{DFT}} - \Omega\boldsymbol{\mathcal{E}} \cdot \mathbf{P}$ at $\boldsymbol{\mathcal{E}} = \pm h_\beta \hat{\mathbf{e}}_\beta$.
2	With ions fixed, evaluate Berry-phase $\mathbf{P}^{\pm\beta}$; unwrap phases and take centered differences to obtain $\boldsymbol{\epsilon}_\infty$.
3	Repeat the same six field calculations with full internal-ion relaxation and self-consistent electrons; use total polarization (electronic plus ionic) to obtain $\boldsymbol{\epsilon}_0$.
4	Compute $\boldsymbol{\epsilon}_{\mathrm{latt}} = \boldsymbol{\epsilon}_0 - \boldsymbol{\epsilon}_\infty$. Optionally cross-check using Born charges and Φ^+ or enthalpy curvature.
5	Validate **k**-mesh, field amplitude, branch tracking, electronic convergence, ionic forces, and crystal symmetries.

Principal final formulas (Gaussian/atomic units):

$$\epsilon^{\infty}_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi \left.\frac{\partial P_\alpha}{\partial \mathcal{E}_\beta}\right|_{\boldsymbol{\mathcal{E}}=0,\,\{\mathbf{R}_I\}\,\mathrm{fixed}},$$

$$\epsilon^{0}_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi \left.\frac{dP_\alpha}{d\mathcal{E}_\beta}\right|_{\boldsymbol{\mathcal{E}}=0,\,\{\mathbf{R}_I\}\,\mathrm{relaxed}},$$

$$\boldsymbol{\epsilon}_{\mathrm{latt}} = \boldsymbol{\epsilon}_0 - \boldsymbol{\epsilon}_\infty.$$

The electronic part of each $\Delta\mathbf{P}$ is available either from occupied-state Berry phases or from the *sum* of occupied Wannier-center shifts. No explicit sum over unoccupied bands is needed in this finite-field approach.

References

References

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